SECTION A:

Item 1

A new taxi park is being constructed in a certain district. The proposed locations for the **Booking Area (T)**, **Waiting Area (A)**, and **Cargo Office (B)** have been surveyed and recorded as:

- Booking Area, **T (2, 4)**
- Waiting Area, **A (5, 10)**
- Cargo Office, **B (8, 16)**

The construction committee will approve the proposed layout only if the three facilities lie on the same straight line.

After the construction is completed, a gate is to be installed at point **D (3,16)**.

The committee wishes to determine the angle $\angle ADT$, as it influences the accessibility of the taxi park to the main road. The proposal will be maintained only if $\angle ADT$ is greater than 270^0

Task

Based on calculations, advise the construction committee whether the proposed taxi park layout should be approved and maintained.

Item 2

A public transport company operates several buses that transport passengers between districts. After every long journey, each bus is taken to the maintenance garage for routine servicing

Engineers have established that **the rate at which the temperature of the engine oil changes is directly proportional to the difference between its temperature and the surrounding temperature in the garage.**

On a particular day, the engine oil has a temperature of **90°C** when the bus arrives at the garage. The garage temperature is maintained at **30°C**.

After **20 minutes**, the temperature of the engine oil has fallen to **60°C**.

According to the company's maintenance policy, servicing should begin only when the engine oil temperature reaches **40°C or below**. If the oil fails to cool to this temperature within **40 minutes**, the company will install an additional cooling system in the maintenance garage.

Task

Based on mathematical justifications, advise the transport company.

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SECTION B:

*This section has **two** parts, **I** and **II***

PART I

*Choose only **one** item*

Item 3

A school library has received a donation of textbooks for different subjects. To make the books easily accessible, the librarian plans to arrange them equally on shelves. If the library uses **t** shelves, the relationship is represented by

$$5 - \frac{6}{t} = t$$

The school administration will only fund library expansion if the number of shelves required is **at most 3**.

To meet the increasing demand for learning resources, the school administration plans to purchase **new textbooks every year**. The number of textbooks purchased each year **doubles** the number purchased in the previous year.

In the **first year**, the administration purchases **4 textbooks**.

The school management intends to continue purchasing textbooks until the **total number of textbooks acquired reaches 508**. However, if it takes **more than 5 years** to reach this target, the management will postpone the expansion of the library because of budget constraints.

To organize the new arrivals, the librarian has selected **six different textbooks** labelled **A, B, C, D, E,** and **F** for display on the **"New Arrivals"** shelf. Due to limited display space, **only four textbooks can be displayed at a time**.

The librarian therefore needs to determine the **number of different ways** in which **four textbooks can be arranged from the six available textbooks**. If the number of possible arrangements **exceeds 300**, the librarian will use a digital system to identify the most effective display arrangement.

Task

Use mathematics justifications to advice;

(a) School administration

(b) Librarian

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Item 4

A community fish farm has recorded a steady increase in the number of fish due to improved feeding practices, better water quality, and effective disease control measures.

At the beginning of last year, the fish farm had **25,000 fish**. Owing to favourable breeding conditions, the fish population is increasing by **4% each year**.

The farm manager plans to **construct an additional fish pond** when the fish population reaches **31,600**.

The farm manager therefore needs to determine **how long it will take for the fish population to reach 31,600** so that some money on the farm budget is diverted to the construction. If the period is less than 7 years, the expansion will start immediately.

A fish farm has six different feed ingredients available for preparing a special feed mixture for young fish.

To produce a balanced diet, the farm manager will mix only three different ingredients at a time. The order in which the ingredients are mixed is important because some ingredients must be added first to improve nutrient absorption, others are added second to balance the feed, while the last ingredient binds the mixture together.

The farm manager therefore needs to determine the number of different ways in which three ingredients can be selected and arranged. From the six available ingredients. If the number of possible arrangements exceeds 100, the farm will hire an expert to determine the most efficient mixing order.

Task

Based on calculations, advise the farm manager.

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PART II

*Choose only **one** item*

Item 5

A farmers' cooperative specializes in the production of **maize** for supply to grain processing industries. Each bag of maize is expected to meet the required quality standards.

To ensure that the harvested maize meets the required production standards, the cooperative's quality assurance team randomly measures the **mass of maize bags** collected from different farms during each harvesting season.

Let x represent the **mass of a randomly selected bag of maize (in tonnes)**. Based on records from previous harvests, the probability density function of x is given by

$$f(x) = \begin{cases} \frac{k}{6}(x+2), & 1 \le x \le 2 \\ 0, & \text{elsewhere} \end{cases}$$

The quality assurance team wishes to determine the **expected mass** of a randomly selected maize bag. The cooperative has suggested that the current production practices will be considered **consistent with the required production standards if the expected mass is at least 1.6 tonnes**.

Recently, the maize farmers' cooperative harvests maize from two farms, Farm A and Farm B. During one harvesting season, the maize is transported to two grain stores, Store X and Store Y.

The quantities of maize delivered (in tonnes) are shown below.

Farm	Store X	Store Y
A	20	15
B	18	22

The cooperative pays the following charges for each tonne of maize delivered to the stores.

Store	Transport Cost (UGX)	Handling Cost (UGX)
X	50,000	5,000
Y	65,000	7,000

'he management wishes to determine the total transport and handling costs for ıaize supplied from each farm. The cooperative has agreed that if the total transport ıd handling cost for any farm exceeds UGX 2,500,000, a maize collection centre 'ill be established near that farm.

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To assess changes in the market value of maize products, a farmers' cooperative collected information on the prices of **three maize products** sold during **2025** and **2026**. The products include **maize grain**, **maize flour**, and **maize bran**.

Maize Product	Maize Grain	Maize Flour	Maize Bran
Price in 2025 (UGX)	2,500	500	1,200
Price in 2026 (UGX)	3,000	500	1,000
Quantity Sold	720	310	505

The cooperative management wishes to determine the **weighted aggregate price index** for **2026**, taking **2025** as the base year, to assess changes in the market prices of maize products.

The management has agreed that **if the weighted aggregate price index for 2026 is less 118, maize production will not be increased during the next planting season.**

Task

Based on statistical evidence, help the;

(a) Quality Assurance team,

(b) Cooperative management

Item 6

A secondary school conducted a **Mathematics test** to evaluate learners' understanding of recently taught concepts. The scores of **learners**, marked out of 100, are shown below.

33, 79, 58, 76, 37, 74, 56, 37, 80, 41

62, 44, 89, 45, 87, 65, 47, 49, 52, 69

87, 49, 54, 77, 86, 57, 61, 58, 65, 30

58, 68, 38, 46, 50, 63, 34, 53, 70, 59

60, 43, 87, 59, 69, 36, 55, 60, 49, 50

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The Head of the Mathematics Department believes that the learners' performance is **reasonably consistent if the interquartile range does not exceed 25 marks**. Otherwise, the department will organize **remedial lessons** to help learners whose performance varies widely.

The school is preparing a team to represent it in an inter-school Mathematics Quiz Competition. Based on records from previous competitions, the Mathematics Department has established that each learner selected for the team has a **0.81 probability** of answering a quiz question correctly. During a practice session, each learner is required to answer **14 independent questions**.

The Head of the Mathematics Department wishes to estimate the **expected number of questions** that a learner is likely to answer correctly. If the expected number of correct answers is **at least 8**, the learner will be considered ready to represent the school in the competition. Otherwise, additional coaching sessions will be organized before the competition.

A secondary school is selecting learners to represent it in a district Mathematics competition. Before selection, each learner is required to complete a **Mathematics test** and a **Mathematics quiz**. The school believes that a learner's performance in the Mathematics test does **not** influence their performance in the Mathematics quiz. From previous school records:

- The probability that a learner passes the **Mathematics test** is **0.8**.
- The probability that the same learner passes the **Mathematics quiz** is **0.7**.

The Head of the Mathematics Department has agreed that a learner will qualify to represent the school **only if the probability of passing both the Mathematics test and the Mathematics quiz exceeds 0.55**.

Task

Based on statistical evidence, Help the head of department.

END